

Research Methodology

Step – 1

Case Studies

HuRoN
15 bags | 16,862 frames
Labels / GT:
- S 15,273: $d \geq 1.2$ m
- P 1,378: $0.6 \leq d < 1.2$ m
- U 211: $d < 0.6$ m

PAL
3 bags | 9,249 frames
Labels / GT:
- S 5,518: $d \geq 0.80$
- P 2,769: $0.35 \leq d < 0.80$ m
- U 962: $d < 0.35$ m

CrowdBot
7 bags | 8,801 frames
Labels / GT:
- S 4,205: $d \geq 1.2$ m
- P 4,350: $0.5 \leq d < 1.2$ m
- U 246: $d < 0.5$ m



Step – 2

VLM as Robot Digital Twin

Visual Language Models

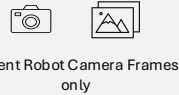
Qwen

InternVL

LLaVA

Approaches

AppFr – Visual History



Recent Robot Camera Frames only

AppOd – Visual History + Context



Recent Robot Camera Frames
Odometry + LiDAR

16 Histories length

Frames Ordered Oldest to newest, last frame = current frame
-- {H2, H4, H6, ..., H32}



Zero-Shot Safety Prompt

Safe | Potentially Unsafe | Unsafe

3 VLMs x 2 Approaches x 16 histories = 96 Configurations

Step – 3

Model Output

Predict Safety label

- Safe + Prob
- Potentially Unsafe + Prob
- Unsafe + Prob



Every Case study has been evaluated independently.

Combined Dataset

25 bags
34,912 frames
Labels / Prediction
- S 24,996
- P 8,497
- U 1,419

Combined Dataset used for Statistical Analysis

Step – 4

Empirical Analysis



RQ1 – Classification Performance

- Primary Metric: Accuracy, Macro F1
- RQ1a: Within Model – Approach Comparison
- RQ1b: Within Approach – Model Comparison
- RQ1c: Overall Model Comparison



RQ2 – Confidence, Uncertainty, Calibration

- Primary Metric: MSP, PCS, DG, ET, ECE, MCE
- RQ2a: Approach Correlation within Model
- RQ2b: Model Correlation within approach
- RQ2c: For Each metric – Model Comparison



RQ3 – Effect of History Length on Performance

- Primary Metric: Accuracy, Macro F1
- History: H2, H4, H6, ..., H32
- RQ1a: History Effect on Accuracy
- RQ1c: History Effect on Macro f1



RQ4 – Early Warning Capability

- Primary Metric: Potentially Unsafe
- Actionable Warning
- Early Warning
- Late Warning
- Missed Warning

Data Collection/Labeling

Experimental Execution

VLM Predictions/Combined

Empirical Analysis {RQ1 – 4}